

NEST: Structural-Entropy Hierarchy Search with General Tree Editing and Exact NNI Certification

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Abstract. Structural entropy evaluates a graph hierarchy through the description length of an encoding tree. Existing methods such as HCSE and BBM construct such trees, but do not determine whether a returned hierarchy still admits an entropy-reducing local topology change. We introduce two complementary tree optimizers. Entropy-Guided Tree Editing (EGTE) constructs a hierarchy from multiple initial trees and monotonically improves the complete objective by level collapse and subtree relocation. NNI-certified Entropy Search over Trees (NEST) replaces this costly general-edit stage with exact rooted nearest-neighbor interchange (NNI). For weighted graphs, we derive its exact entropy change and prove that an improving NNI can be inaccessible to a monotone single merge or compression. The identity supports best-improvement descent, a checkable one-NNI local-optimality certificate, and a bounded two-move search that can cross a controlled intermediate barrier while committing only an entropy-decreasing endpoint. All synthetic comparisons are paired by graph seed and use the same independently recomputed objective. Across 50 graphs from five hierarchical regimes, NEST attains the lowest mean entropy in every regime. Relative to the better of HCSE and BBM on each graph, it lowers entropy by 0.4290 ± 0.0281 bits and wins all 50 paired comparisons. As an internal efficiency comparison, NEST also lowers entropy by 0.0279 ± 0.0072 bits while running $21.5\times$ faster than EGTE under the instrumented benchmark.

Keywords: Structural entropy · hierarchical clustering · tree editing · nearest-neighbor interchange · graph algorithms

1 Introduction

Many networks are organized at more than one scale: vertices form local groups, related groups form larger modules, and the desired output is therefore a tree rather than one flat partition. Structural entropy supplies an information-theoretic objective for such an encoding tree [8]. Recent HCSE and BBM methods construct low-entropy hierarchies through recursive partitioning and balanced merging [12]. Their construction rules, however, do not answer a complementary question: *after a tree has been constructed, does a small topology change still lower the same objective?*

Construction and local optimality are distinct: early decisions can retain avoidable entropy, while unrestricted tree editing considers many source–destination pairs and repeatedly rescores the full objective. We seek both broad construction and a local operator with an exact effect and interpretable termination.

Entropy-Guided Tree Editing (EGTE) starts from SE agglomeration, recursive SE, and optionally Paris, then applies exact-guarded collapse and relocation. This label-free general search is strong but costly because it repeatedly rescores the whole tree, motivating a cheaper certifiable neighborhood.

Nearest-neighbor interchange (NNI) is the elementary rotation of a rooted binary tree. It changes one internal edge while preserving every leaf. NNI is standard in phylogenetic tree search and has been studied for other hierarchical-clustering objectives, but its structural-entropy change has not been used as an exact graph-hierarchy optimizer. We show that, for $((A, B), C) \rightarrow (A, (B, C))$, all objective terms cancel except two cross-module edge weights and three module volumes. This identity yields monotone descent and a one-NNI local-optimality certificate.

Our full fast method, NNI-certified Entropy Search over Trees (NEST), refines several inexpensive initial trees with the exact operator and returns the lowest independently verified endpoint. A bounded two-move beam can cross a small intermediate barrier, but accepts a pair only when its endpoint lowers the full objective. This design retains exact acceptance while avoiding the costliest part of general tree editing.

Our contributions are:

- EGTE, a label-free multi-start hierarchy optimizer with exact-guarded collapse and subtree-relocation moves;
- an exact weighted-graph NNI delta, a proof of separation from monotone single merge/compress steps, and a checkable one-NNI local certificate;
- NEST, which combines multi-start construction, exact NNI refinement, and bounded two-move escape without weakening the output guarantee; and
- a paired evaluation against SE agglomeration, Louvain-2L, Paris, and the official HCSE and BBM implementations, with entropy, hierarchy recovery, runtime, memory, operator audits, and an explicit EGTE–NEST ablation.

Across five frozen regimes, NEST lowers entropy by 0.5512 ± 0.0632 bits versus HCSE and 0.5175 ± 0.0319 bits versus BBM, winning every paired comparison. It provides a fast locally certified alternative to general editing.

2 Preliminaries

Weighted graph and encoding tree. Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights. For $S \subseteq V$, let $V_S = \text{vol}(S) = \sum_{u \in S} d_u$, and let g_S be the total weight of edges with exactly one endpoint in S . The total graph volume is $\text{vol}(V) = \sum_u d_u = 2 \sum_{e \in E} w_e$. An encoding tree T has one leaf per vertex. Each internal node a represents the union S_a of its descendant leaves; a^- denotes its parent.

The structural entropy of G under T is

$$H^T(G) = - \sum_{a \neq r} \frac{gs_a}{\text{vol}(V)} \log_2 \frac{V_{S_a}}{V_{S_a^-}}. \quad (1)$$

Our sole optimization objective is Eq. (1); lower is better. No ground-truth label is used to construct or select a tree. For a zero-volume module, its summand is defined as zero; such a module has no incident positive-weight edge and can be attached after optimizing the positive-volume part of the tree.

Constructors. We distinguish a *constructor*, which produces an initial tree, from a *refiner*, which searches neighboring topologies. Our candidate pool contains (i) a binary SE-agglomerative dendrogram, (ii) a recursive SE partition tree, and (iii) a Paris dendrogram when the optional implementation is available [1]. HCSE and BBM are distinct external constructors and are evaluated from their official implementation [12].

Rooted NNI. An NNI move acts on a binary parent-child edge. Writing the local topology as $((A, B), C)$, the two nontrivial alternatives are $(A, (B, C))$ and $(B, (A, C))$. The leaf set, graph, and all modules outside the local parent $P = A \cup B \cup C$ remain unchanged. A rooted binary tree with n labeled leaves has finitely many topologies, namely $(2n - 3)!!$, so any strictly decreasing local search terminates.

3 Entropy-Guided Tree Search

Our framework contains two complementary optimizers. EGTE searches a broad space of non-binary hierarchy edits and provides a strong general solution. NEST uses the exact local identity developed below to obtain a substantially faster hierarchy with an explicit NNI-local certificate.

3.1 Entropy-guided general tree editing

EGTE first constructs a pool containing an SE-agglomerative dendrogram, a recursive-SE partition tree, and a Paris tree when available. For each start it searches two topology-changing neighborhoods. A *collapse* removes an internal level and connects its children directly to its parent. A *relocation* detaches a subtree and attaches it to a nearby internal module, allowing information from one branch to reorganize another. Every candidate tree is evaluated with the complete objective in Eq. (1); a move is accepted only when this independently recomputed value strictly decreases. Small graphs use best improvement over all candidates, while larger graphs use shuffled first improvement with a declared relocation cap.

Proposition 1 (Monotonicity of general tree editing). *The entropy returned by EGTE is no greater than that of the best successfully constructed initial tree.*

Proof. Each edit is committed only after exact recomputation verifies a strict decrease, and EGTE returns the lowest-entropy refined candidate.

Collapse and relocation provide a broad edit space, but repeated full-tree evaluation is costly and their termination does not characterize a canonical binary neighborhood. We therefore develop an exact NNI refiner whose local effect can be evaluated without rescoring every unaffected module.

3.2 A local structural-entropy identity

For disjoint sets X, Y , write $W(X, Y) = \sum_{u \in X, v \in Y} w_{uv}$. Consider the rotation shown in Fig. 1. Let $X = A \cup B$, $Y = B \cup C$, and $P = A \cup B \cup C$.

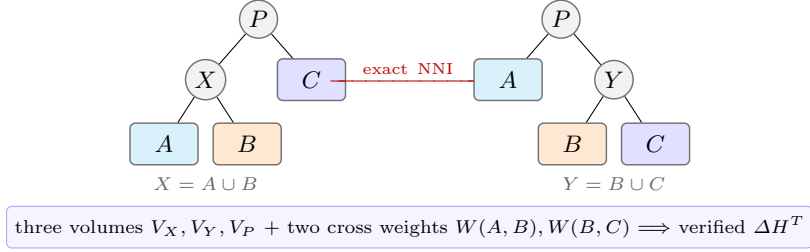


Fig. 1. A rooted NNI replaces only $X = A \cup B$ by $Y = B \cup C$. The highlighted local statistics determine the exact full-objective change; all leaves and the surrounding tree remain fixed.

Theorem 1 (Exact weighted NNI delta). *For an eligible rotation with $V_X, V_Y, V_P > 0$, the change in tree structural entropy is*

$$\Delta H^T = \frac{2}{\text{vol}(V)} \left[W(A, B) \log_2 \frac{V_P}{V_X} + W(B, C) \log_2 \frac{V_Y}{V_P} \right]. \quad (2)$$

The symmetric alternative follows by exchanging A and B .

Proof. First suppose the three child modules have positive volume; the zero-volume boundary case follows by deleting modules with no incident positive-weight edge and taking the corresponding zero-contribution limit. Only the terms for A, B, C, X , and Y can change. Omitting the common factor $1/\text{vol}(V)$, their old and new contributions are

$$\begin{aligned} L_{\text{old}} &= g_A \log_2 \frac{V_X}{V_A} + g_B \log_2 \frac{V_X}{V_B} + g_X \log_2 \frac{V_P}{V_X} + g_C \log_2 \frac{V_P}{V_C}, \\ L_{\text{new}} &= g_A \log_2 \frac{V_P}{V_A} + g_B \log_2 \frac{V_Y}{V_B} + g_C \log_2 \frac{V_Y}{V_C} + g_Y \log_2 \frac{V_P}{V_Y}. \end{aligned}$$

Use $g_X = g_A + g_B - 2W(A, B)$ and $g_Y = g_B + g_C - 2W(B, C)$. In $L_{\text{new}} - L_{\text{old}}$, the coefficients of g_A, g_B, g_C telescope to zero. The remaining terms are

$$2W(A, B) \log_2(V_P/V_X) + 2W(B, C) \log_2(V_Y/V_P).$$

Restoring $1/\text{vol}(V)$ proves Eq. (2). In particular, external boundary weights and $W(A, C)$ do not enter the rotation delta.

Proposition 2 (Atomic advantage over merge-compress). *Let T_c contract the edge from P to $X = A \cup B$, exposing A, B, C as siblings, and let T' then merge B, C under $Y = B \cup C$. For these HCSE compress and merge primitives,*

$$H^{T_c}(G) - H^T(G) = \frac{2W(A, B)}{\text{vol}(V)} \log_2 \frac{V_P}{V_X} \geq 0, \quad (3)$$

$$H^{T'}(G) - H^{T_c}(G) = -\frac{2W(B, C)}{\text{vol}(V)} \log_2 \frac{V_P}{V_Y} \leq 0. \quad (4)$$

Their sum is Eq. (2). Thus NNI lowers entropy when the merge gain in Eq. (4) exceeds the compression loss in Eq. (3), although neither desired reparenting nor an improving compression is available as one primitive.

Proof. Delete X 's codeword and substitute $g_X = g_A + g_B - 2W(A, B)$ in Eq. (1); this gives Eq. (3). The symmetric calculation for inserting Y gives Eq. (4).

This proves strict separation from monotone single-primitive search, not global dominance over a complete HCSE round, which may cross an intermediate increase. We evaluate Eq. (2) from cached sets and sparse adjacency lists, then verify every selected move by fully recomputing Eq. (1).

3.3 One-step certificate and compound escape

Algorithm 1: Exact NNI refinement of one candidate tree

Input: Graph G , tree T , tolerance ε , barrier β , beam b

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1 repeat
2   | evaluate Eq. (2) for every legal NNI of  $T$ ;
3   | apply the most negative move and verify the full objective;
4 until no improving one-step move exists;
5 for at most  $R$  compound rounds do
6   | retain the  $b$  first moves with smallest  $\Delta H$  and  $\Delta H \leq \beta$ ;
7   | expand every legal second NNI from each retained intermediate tree;
8   | if no two-move endpoint has entropy below  $H^T(G)$  then
9     |   break;
10  | commit the best endpoint; rerun one-step descent;
11 return  $T$ 
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Proposition 3 (Monotonicity and local certificate). *One-step refinement never increases $H^T(G)$. If it stops before its move budget, the returned tree has no improving rooted NNI on any eligible binary parent-child edge.*

Proof. Only moves with $\Delta H^T < -\varepsilon$ are committed, and their endpoint is independently verified. At termination every enumerated legal move has $\Delta H^T \geq -\varepsilon$.

Strict one-step descent can stop before a better tree separated by a shallow barrier. Algorithm 1 therefore keeps a bounded beam of first moves whose increase is at most β and evaluates a second NNI. It never commits the intermediate tree alone.

Proposition 4 (Safe compound search). *The compound stage never worsens its input. It can reach an improving endpoint even when every one-step neighbor is non-improving.*

Proof. A pair is committed only after the final tree is fully recomputed and found to have lower entropy than the tree preceding the pair. The first move is not committed independently, so its sign cannot weaken the output guarantee. A two-edge path in tree space may have a positive first delta and a negative net delta; the regression witness in Sec. 4 instantiates this case.

3.4 Fast NNI-certified entropy search

NEST applies Algorithm 1 independently to SE-agglomerative, recursive-SE, and Paris starts and returns the candidate with minimum verified entropy. Unlike EGTE, it omits general collapse/relocation search and uses the exact NNI neighborhood directly. The initializer is recorded for every output, so gains can be attributed to candidate selection, one-step refinement, and compound escape. With n leaves there are at most $2(n-2)$ oriented NNI alternatives per sweep, versus the larger source-destination edit space of EGTE. Our implementation uses best improvement, cached leaf sets, sparse cross-weight scans, and an $O(m \log h + n)$ full verification only for a selected move, where h is tree height. We make runtime claims empirically rather than assuming balanced trees.

4 Experiments

4.1 Questions and protocol

We ask: (Q1) how often do established constructors leave an improving NNI; (Q2) does compound search add improvements beyond a one-step certificate; (Q3) does lower entropy preserve or improve recovery of a planted hierarchy; and (Q4) how do EGTE and NEST compare with HCSE and BBM, and what efficiency is gained by replacing general editing with exact NNI?

The frozen protocol uses ten graph seeds per regime. Every method sees the identical weighted graph, and every tree is rescored by the same independent

implementation of Eq. (1). We report paired means and 95% t -intervals, raw per-seed records, wall-clock time, and Python allocation peak. The compound search uses beam width 16, barrier $\beta = 0.05$ bits, and at most eight accepted compound rounds. No label is used by EGTE or NEST for model selection. BBM is additionally shown with the planted number of fine clusters, which is an oracle advantage and is marked as such.

Data. Our controlled suite contains five two-level hierarchical stochastic block models: clean, noisy, imbalanced, weighted, and weak-hierarchy. Each balanced graph has 64 vertices, eight fine blocks, and four coarse blocks; the imbalanced regime has the same total size with block sizes from 4 to 12. The generator manifest, any 0.01-weight connectivity edges, and every seed are stored with the results. This family follows the hierarchical stochastic block model used to study recovery beyond worst-case inputs [4].

Methods. We compare SE agglomeration, a two-level Louvain tree, Paris [1], official HCSE, official BBM with oracle k [12], and our EGTE and NEST variants. The proposed methods are separated from external baselines in every table. We also refine each constructor independently to measure its one-step and compound gap rather than crediting all gains to our preferred initializer.

Metrics. The primary metric is $H^T(G)$. Dendrogram purity is computed for the planted fine and coarse partitions: for every same-class vertex pair, it measures the class fraction under the pair’s predicted LCA, then averages over pairs. We also report the fraction of runs improved by one-step and compound NNI, number of moves, runtime, and memory.

4.2 A strict one-step trap

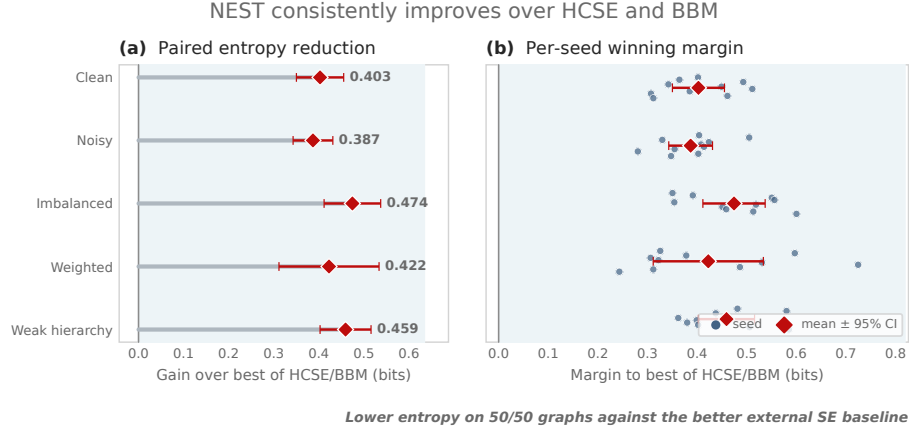
Before aggregate evaluation, an eight-vertex weighted witness verifies the purpose of compound search. Exact one-step descent stops at 1.970038 bits. Every one-NNI delta is nonnegative, but a first move crossing a 0.013970-bit barrier followed by a second move and ordinary descent reaches 1.920593 bits, a 0.049445-bit reduction. This witness is generated from a fixed seed and is covered by a regression test.

4.3 Frozen benchmark results

Table 1 reports the raw output of each method, before applying our audit to competing trees. NEST has the lowest mean H^T in all five regimes and on 50/50 individual paired graphs. Its paired reduction is 0.5512 ± 0.0632 bits relative to HCSE and 0.5175 ± 0.0319 bits relative to BBM; it is lower on all 50 graphs against each method. As an internal comparison, NEST improves over EGTE by 0.0279 ± 0.0072 bits, while fine-level dendrogram purity changes by $+0.016 \pm 0.010$. Thus the objective gain is not obtained by systematically degrading planted recovery. Figure 2 shows paired evidence against the stronger of HCSE and BBM on each graph.

Table 1. Tree structural entropy H^T (mean $\pm$ 95% CI; lower is better). $\text{BBM}^\dagger$ receives the planted fine-cluster count.

Method	Clean	Noisy	Imbal.	Weighted	Weak
SE-agglomerative	2.920 ± 0.094	3.712 ± 0.070	3.218 ± 0.116	2.444 ± 0.079	3.424 ± 0.112
Louvain-2L	4.167 ± 0.066	4.575 ± 0.061	4.270 ± 0.060	3.647 ± 0.145	4.425 ± 0.095
Paris	2.919 ± 0.112	3.698 ± 0.072	3.165 ± 0.106	2.348 ± 0.072	3.433 ± 0.112
HCSE	3.300 ± 0.190	4.351 ± 0.170	3.685 ± 0.122	2.759 ± 0.130	4.040 ± 0.286
$\text{BBM}^\dagger$	3.359 ± 0.122	4.062 ± 0.073	3.634 ± 0.124	2.996 ± 0.073	3.916 ± 0.064
EGTE (ours)	2.903 ± 0.103	3.691 ± 0.071	3.164 ± 0.106	2.348 ± 0.072	3.413 ± 0.110
NEST (ours)	2.845 ± 0.094	3.675 ± 0.067	3.135 ± 0.115	2.337 ± 0.073	3.387 ± 0.104

**Fig. 2.** Paired advantage over external structural-entropy methods. For each graph, the reference is the lower-entropy result of HCSE and BBM. (a) Regime-level mean reduction and 95% paired confidence interval. (b) Per-seed margins with small vertical jitter for visibility. NEST has lower entropy on all 50 graphs.

4.4 The operator audit

The constructor audit in Table 2 asks whether a returned tree can be improved by the exact local operator. One-step NNI improves SE agglomeration and oracle BBM on every graph and improves the proposed EGTE output on 80% of graphs. After one-step descent has certified its local neighborhood, bounded compound search still improves 72% of SE-agglomerative trees, 92% of BBM trees, and 40% of EGTE trees. Paris is closer to local optimality but is not immune. For HCSE, one-step NNI lowers the imbalanced seed-5 tree by 0.001352 bits, while compound search lowers the weighted seed-2 tree by 0.000637 bits. These explicit witnesses support the neighborhood-separation claim without asserting that HCSE is generally weak.

Table 2. NNI audit pooled over 50 paired graphs. Reduction is relative to each constructor’s raw H^T ; rates are the fractions of runs improved.

Constructor	1-NNI gain	1-NNI hit	2-step gain	2-step hit	Time
SE-agglomerative	1.99%	100%	0.15%	72%	0.334s
Paris	0.06%	68%	0.07%	38%	0.181s
HCSE	0.00%	2%	0.00%	2%	0.055s
BBM	13.98%	100%	0.47%	92%	1.038s
EGTE (ours)	0.36%	80%	0.06%	40%	0.200s

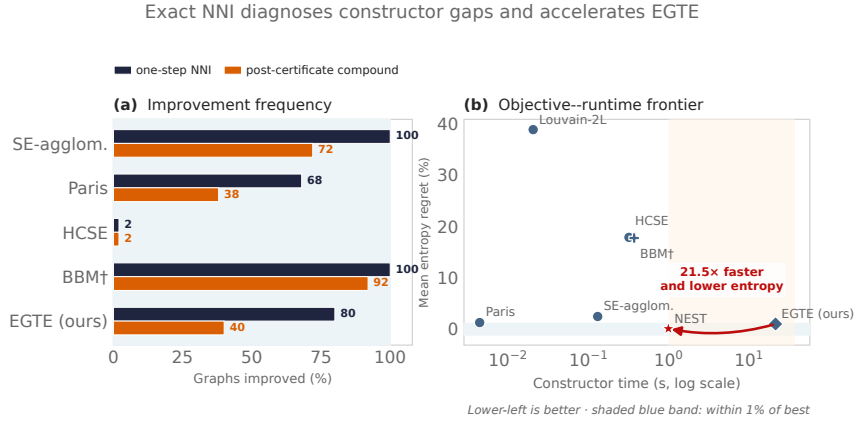


Fig. 3. Operator diagnostics pooled over 50 paired graphs. (a) Fraction of graphs improved by one-step NNI and by compound search after the one-step certificate. (b) Mean entropy regret versus runtime; lower-left is better, and the arrow compares the proposed EGTE and NEST variants. Runtime includes Python allocation tracing and should be interpreted comparatively.

4.5 Interpretation and reproducibility

NEST is not merely EGTE followed by rotations. It forms a small candidate set from SE agglomeration, recursive SE, and Paris, skips general editing, refines each candidate with the same exact operator, and returns the lowest independently rescored endpoint. The benchmark contains 350 method records and 50 generator manifests. Its verifier checks schema, paired seeds, unique keys, and monotonicity of committed NNI endpoints. JSON evidence, scripts, and the HCSE/BBM commit accompany the paper.

Table 3 separates the three algorithmic choices. The candidate pool improves 34 of 50 graphs over SE agglomeration; exact one-step NNI then improves 45 of 50 paired multi-start outputs; bounded compound search adds a smaller but nonzero gain on 30 graphs. Hence neither the initializer pool nor the escape stage alone explains the full result.

Table 3. Component ablation pooled over 50 paired graphs. The gain and improved-run columns compare each row to the preceding row.

Variant	Mean H^T	Mean gain	Improved runs
SE agglomeration	3.1435	–	–
+ candidate pool	3.1041	0.0394	34/50
+ exact 1-NNI	3.0794	0.0247	45/50
+ compound escape	3.0759	0.0035	30/50

Table 4. Tree entropy on four bundled real networks (lower is better). BBM uses the ground-truth k when labels exist and Louvain’s k otherwise.

Method	Karate	Florent.	Les Mis.	Davis
SE-agglomerative	2.698	1.923	3.078	3.055
Louvain-2L	3.405	2.452	3.923	3.813
Paris	2.597	1.923	2.920	2.981
HCSE	2.929	2.172	3.930	3.206
BBM	3.330	2.254	3.199	3.387
EGTE (ours)	2.597	1.923	2.920	2.981
NEST (ours)	2.586	1.919	2.916	2.962

4.6 Real networks and clean scaling

Table 4 complements the controlled suite with four networks bundled by NetworkX or the artifact library. NEST obtains the lowest raw entropy on 4/4 networks. Because only Karate has a standard class label in the bundled loader and the remaining networks lack a unique ground-truth hierarchy, this experiment tests objective minimization rather than recovery.

Table 5 removes allocation tracing and varies graph size. NEST lowers entropy in 12/12 paired runs. Its mean speedup over EGTE grows from $9.7\times$ at 32 vertices to $22.2\times$ at 128 vertices. These graphs remain small enough for exact rescaling; the trend supports efficiency, not an asymptotic complexity claim.

5 Related Work

Structural entropy and hierarchy construction. Li and Pan introduced structural information as an encoding-tree measure of network organization [8]. HCSE and BBM develop hierarchy construction and balancing operations for this objective [12]. The structural-entropy survey organizes the broader theory, optimization, and application literature [14]; HypCSE subsequently develops a continuous hyperbolic formulation for hierarchical clustering [15]. EGTE complements these constructors with exact-guarded general tree editing; NEST then asks whether any returned tree is locally optimal under an exact topology rotation and uses the answer as both a diagnostic and an optimizer.

Table 5. Clean wall-clock scaling without allocation tracing; means over three graph seeds. EGTE and NEST are our general-edit and certified fast variants.

n	H^T		Time (s)		Speedup
	EGTE	NEST	EGTE	NEST	
32	2.485	2.453	0.430	0.044	9.7×
64	3.140	3.098	2.941	0.188	15.7×
96	3.491	3.369	9.717	0.482	20.2×
128	3.746	3.623	17.962	0.811	22.2×

Graph hierarchical clustering. Classical hierarchical clustering is predominantly agglomerative or divisive [11]. Paris is an agglomerative graph hierarchy related to modularity and provides a strong inexpensive initialization [1]. Dasgupta-style objectives initiated a complementary algorithmic literature [6]; average linkage and divisive local search have guarantees for its revenue dual [10]. Sparsest-cut and spreading-metric algorithms give approximations for that family [2, 13], while admissible hierarchical objectives have been characterized more generally [5]. Continuous ultrametric fitting offers another optimization route [3]. These objectives are useful conceptual neighbors, but a local delta derived for one cannot be substituted for Eq. (1).

Local search in tree space. NNI is a classical neighborhood for phylogenetic trees; its reconfiguration distance is itself nontrivial [9]. Recent work studies NNI local search for hierarchical-clustering costs [7]. Our formula targets weighted graph structural entropy and exposes the two cross-module weights controlling a rotation. Unlike unit-clique landscape analyses, our method evaluates the complete objective on arbitrary weighted graphs. Its new content is the verified optimizer, compound search, multi-start construction, and comparative evaluation; local certification is not global optimality.

6 Limitations and Conclusion

NNI is a local operator, not a certificate of global optimality. Compound search explores only a bounded two-step neighborhood, while EGTE uses a finite edit budget; neither exhausts all encoding trees. Multiway regions must also be resolved by a constructor before binary rotations can act on them. Lower structural entropy need not improve every external notion of hierarchy, which is why we report planted purity alongside the optimized objective.

We introduced two structural-entropy hierarchy optimizers. EGTE performs exact-guarded level collapse and subtree relocation over multiple initial trees. NEST derives an exact weighted NNI identity and uses it for fast monotone refinement, a local-optimality certificate, and safe bounded escape from shallow one-step traps. On the frozen five-regime suite, NEST achieved the lowest mean entropy in every regime and outperformed both HCSE and BBM on all 50

paired graphs. The operator audit further shows that hierarchy construction and local tree optimality are distinct concerns. Together, EGTE and NEST turn a broad but costly edit strategy into a reproducible global–local framework whose guarantees and empirical gains can be checked directly.

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